

Assignment_Equations_Matrices

1. Solve $AX=0$ where $A = \begin{pmatrix} 2 & -1 & 3 & 2 \\ 1 & 4 & 0 & -1 \\ 2 & 6 & -1 & 5 \end{pmatrix}$.

2. Solve

$$\begin{aligned} x_1 - 2x_2 + x_3 + 2x_4 &= 1 \\ x_1 + x_2 - x_3 + x_4 &= 2 \\ x_1 + 7x_2 - 5x_3 - x_4 &= 3 \end{aligned}$$

3. Find all solutions of

$$\begin{aligned} 2x_1 - 3x_2 - 7x_3 + 5x_4 + 2x_5 &= -2 \\ x_1 - 2x_2 - 4x_3 + 3x_4 + x_5 &= -2 \\ 2x_1 - 4x_3 + 2x_4 + x_5 &= 3 \\ x_1 - 5x_2 - 7x_3 + 6x_4 + 2x_5 &= 7 \end{aligned}$$

4. Let $A = \begin{pmatrix} 3 & -6 & 2 & -1 \\ -2 & 4 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 1 & -2 & 1 & 0 \end{pmatrix}$. For which $b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}$ such that $AX=b$ has a solution?

5. Show that $\begin{pmatrix} 1 & 1 & 2 \\ -2 & 0 & -1 \\ 1 & 3 & 5 \end{pmatrix}$ is not row equivalent to $\begin{pmatrix} 2 & 0 & 0 \\ a & -1 & 0 \\ b & c & 3 \end{pmatrix}$ for any a, b, c .

6. Let A, B be two $n \times n$ matrices. Show that the sum of diagonal elements of AB is equal to that of BA .

7. Given $A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 0 & 1 \\ 3 & 0 & 1 \end{pmatrix}$, find elementary matrices $E_k, \dots, E_1$ such that

$$E_k \cdots E_1 A = I$$

8. Let $A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ -1 & 0 & 3 & 5 \\ 1 & -2 & 1 & 1 \end{pmatrix}$. Find an invertible matrix Q such that QA is in row reduced echelon form.

9. Suppose A, B are $m \times n$ matrices such that $AX = 0$ and $BX = 0$ have the same set of solutions. Show that there exists $m \times m$ invertible matrix Q such that $QA = B$.

10. For each of the following matrices $\begin{pmatrix} 2 & 5 & -1 \\ 4 & -1 & 2 \\ 6 & 4 & 1 \end{pmatrix}, \begin{pmatrix} 1 & -1 & 2 \\ 3 & 2 & 4 \\ 0 & 1 & -2 \end{pmatrix}$, find the

inverse matrix if exists.

11. Let A be an $n \times n$ matrix. Show that A is invertible iff $AX = 0$ has only trivial solution.
12. Let A be an $n \times n$ matrix. Show that A is invertible iff $AX = b$ is solvable for any b .
13. Let a be an $n \times 1$ matrix and b be a $1 \times n$ matrix. Let $A = ab$. Show that A is not invertible.
14. Prove a generalization of the previous exercise. If A is an $m \times n$ matrix and B is an $n \times m$ matrix, then AB is not invertible when $m > n$.

15. Show that $A = \begin{pmatrix} 1 & \frac{1}{2} & \cdots & \frac{1}{n} \\ \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n+1} \\ \vdots & \vdots & & \vdots \\ \frac{1}{n} & \frac{1}{n+1} & \cdots & \frac{1}{2n-1} \end{pmatrix}$ is invertible.

16. Find the LU decomposition for

$$\begin{pmatrix} 6 & 18 & 3 \\ 2 & 12 & 1 \\ 4 & 15 & 3 \end{pmatrix}$$

17. (Not for submission) Suppose A is an invertible matrix. Can you use our current techniques to show that a matrix B sufficiently close to A is also invertible?